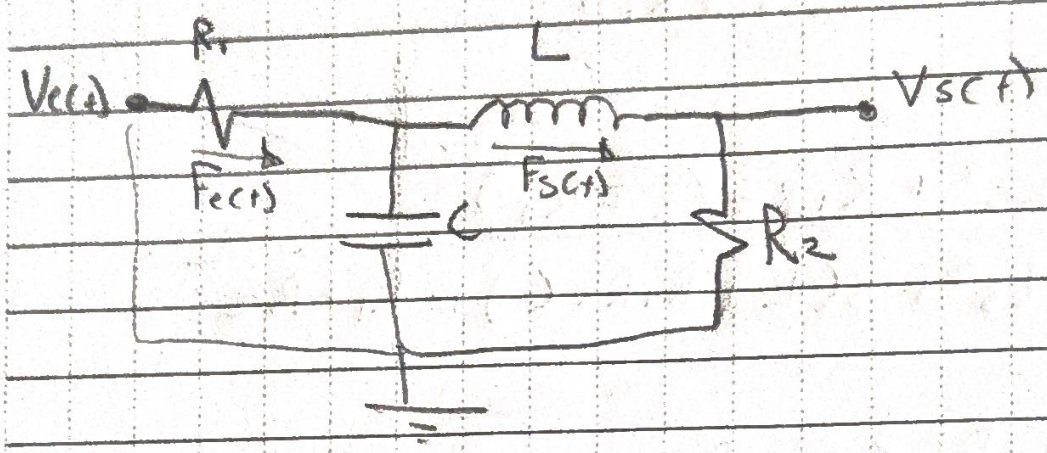
Date / /



$$V_e(t) = R_1 F_e(t) + \frac{1}{C}\int [F_e(t) - F_s(t)]\,dt$$

$$V_s(t) = R_2 F_s(t)$$

$$\frac{1}{C}\int [F_e(t) - F_s(t)]\,dt = \left[\frac{d F_s(t)}{dt} + R_2 F_s(t)\right]$$

<u>Transformada de Laplace</u>

$$V_e(s) = R_1 F_e(s) + \frac{F_e(s) - F_s(s)}{Cs}$$

$$V_s(s) = R_2 F_s(s)$$

$$\frac{F_e(s) - F_s(s)}{Cs} = L_s F_s(s) + R_2 F_s(s)$$

<u>Procedimiento algebraico</u>

$$\frac{F_e(s)}{Cs} = F_s(s)\left(Ls + R_2 + \frac{1}{Cs}\right)$$

$$F_e(s) = F_s(s)\,\cancel{Cs}\left(\frac{CLs^2 + R_2 Cs + 1}{\cancel{Cs}}\right)$$

$$F_e(s) = F_s(s)\left(CLs^2 + R_2 Cs + 1\right)$$

estrella-x

Date / /

$$V_e(s) = F_c(s)\left(R_1 + \frac{1}{Cs}\right) - F_s(s)\left(\frac{1}{Cs}\right)$$

$$= F_s(s)\left[\left(CLs^2 + R_2Cs + 1\right)\left(\frac{R_1Cs+1}{Cs}\right) - \frac{1}{Cs}\right]$$

$$= F_s(s)\left[\frac{R_1C^2Ls^3 + CLs^2 + R_1R_2C^2s^2 + R_2Cs + R_1Cs + \cancel{1} - \cancel{1}}{Cs}\right]$$

$$= F_s(s)\left[\frac{\cancel{Cs}\left(R_1CLs^2 + Ls + R_1R_2Cs + R_2 + R_1\right)}{\cancel{Cs}}\right]$$

$$V_e(s) = F_s(s)\left(R_1CLs^2 + s\left(L + R_1R_2C\right) + R_1 + R_2\right)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\cancel{F_s(s)}\,(R_2)}{\cancel{F_s(s)}\left(R_1CLs^2 + s\left(L + R_1R_2C\right) + R_1 + R_2\right)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{R_2}{R_1CLs^2 + s\left(L + R_1R_2C\right) + R_1 + R_2}$$

<u>Estabilidad en lazo abierto</u>

$$R_1CLs^2 + s\left(L + R_1R_2C\right) + R_1 + R_2$$

$$\lambda = \frac{-L - R_1R_2C \pm \sqrt{\left(L + R_1R_2C\right)^2 - 4R_1CL\left(R_1 + R_2\right)}}{2R_1CL}$$

λ_1

ESTRELLA-X

$$E(s) = \lim_{s \to 0} \left[s \times \frac{1}{s} \left(1 - \frac{R_2}{R_1 C L s^2 + (L + R_1 R_2 C) s + R_1 + R_2} \right) \right]$$

$$E(s) = 1 - \frac{R_2}{R_1 + R_2}$$

$$E(s) = \frac{R_1 + \cancel{R_2} - \cancel{R_2}}{R_1 + R_2}$$

$$E(s) = \frac{R_1}{R_1 + R_2}$$

Caso: $R_1 = 1k$, $R_2 = 1k$

$$E(s) = \frac{1k}{1k + 1k} = 0.5$$

Control

$R_1 = 1k$ $R_2 = 100k$

$$E(s) = \frac{1k}{1k + 100k} = 0.0099$$

```
Lambda 1 Control: -999989.9898896872
Lambda 2 Control: -1010.0101103126214

Función de transferencia del sistema: <TransferFunction>: sys[41]
Inputs (1): ['u[0]']
Outputs (1): ['y[0]']

           1000
  -----------------------
  0.1 s^2 + 1000 s + 2000

Lambda 1 Caso: -9998.999799939978
Lambda 2 Caso: -2.000200060022009
```